

Howard (Hao) Ye

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[LinkedIn](#) — [GitHub](#) — [Portfolio](#)

SUMMARY

Software Engineer with 8+ years of experience designing and deploying distributed systems, full-stack web applications, and applied machine learning solutions. Expert in bridging complex data workflows with secure cloud infrastructure to build interactive, production-grade platforms for healthcare, energy optimization, and advanced analytics.

SKILLS

Backend & Platform: Node.js, Express, Go ([gorilla/mux](#)), Python, FastAPI, TypeScript, REST APIs

AI & Automation: Google Gemini API, Reinforcement Learning (PPO), prompt engineering, [@atproto/api](#), context pipelines

DevOps & Cloud: Docker, Kubernetes, GitHub Actions, Jenkins, Ansible, Terraform, GCP, AWS

Data & ML: Microsoft Fabric, PySpark, SQL, PostgreSQL, SQLite, Redis, MLflow, Scikit-learn, Data Vault 2.0

Frontend & Mobile: React, React Native (Expo), Vue.js, Tailwind CSS, JavaScript, HTML/CSS, Socket.IO

Security & Observability: JWT, RBAC, Prometheus, Grafana, fail2ban, WireGuard, rate limiting, TLS

Languages: Python, TypeScript, JavaScript, Go, Bash, Java, C/C++

EDUCATION

Master of Science in Computer Engineering (Research Track) — University of Tennessee Knoxville
2016–2025

EXPERIENCE

Graduate Research Assistant — **University of Tennessee Knoxville** **2017–2025**

Research Focus: HPC Systems Performance Modeling & Federated Learning for Energy Optimization

- Extracted and processed IOR benchmark telemetry across sequential, random, and strided workload patterns on a production Lustre HPC cluster, utilizing the data to parameterize independent performance models.
- Engineered a discrete-event simulator in OMNeT++ (C++) and a Colored Petri Net model to predict throughput, server utilization, and sojourn times, demonstrating familiarity with HPC batch job scheduling.
- Analyzed real-world Lustre OSS/OST telemetry to identify data quality limitations, evaluating noisy server-side logs to establish critical constraints for ground-truth validation and future cluster scaling.
- Architected a federated learning system for smart-home energy optimization, enabling localized model training on appliance usage to dynamically output cost-minimizing schedules based on grid pricing and solar/battery states.

AI Software Engineer — **MatchingDonors Inc** — [\[Live Platform\]](#) — [\[Web Repo\]](#) — [\[Agent Repo\]](#) **Jan 2026–Present**

- Architected an AI-integrated, full-stack monorepo matching platform to optimize resource discovery for patients, donors, sponsors, and admins, deploying the Node.js/Express backend to Render and the React frontend to Vercel.
- Engineered strict Role-Based Access Control (RBAC) and role-specific workflows across a React web portal and a React Native (Expo) mobile app, securing stateless sessions via JWT and device keystores (SecureStore).
- Developed an autonomous, cron-driven AI Social Media Agent (TypeScript/Express) to systematically scrape medical data, leveraging Google Gemini to draft, auto-condense, and publish optimized social threads.
- Designed a "Human-in-the-Loop" (HitL) React dashboard utilizing strict mutex locks and synchronized "ghost timers," allowing administrators to safely interrupt automated AI scraping pipelines to perform manual interventions.
- Orchestrated the AI agent's infrastructure via Docker Compose and deployed a highly available, 5-node replica swarm utilizing Kubernetes (K8s) and a centralized PostgreSQL database.

- Established dual CI/CD pipelines utilizing GitHub Actions and an on-premise Dockerized Jenkins environment, validating all systems through comprehensive Jest and Supertest integration suites.

PROJECTS

Insurance Fraud Detection Platform on Microsoft Fabric — [\[Repo\]](#)

- Architected an end-to-end insurance fraud analytics platform on Microsoft Fabric, transforming raw claims data into actionable risk signals for downstream investigation.
- Engineered a Bronze/Silver/Gold medallion data pipeline using PySpark and SQL Analytics Endpoints to ingest, clean, and extract features from 15,420 vehicle claims.
- Managed model lifecycle and evaluation using MLflow and Spark MLlib, generating curated Delta tables to ensure features and scored outputs are reusable for reporting.
- Addressed class imbalance by fine-tuning probability thresholds across multiple classifiers, with the leading Random Forest model achieving an AUC of 0.94 and an F1 score of 0.87 on held-out data.

Insurance Claims Data Vault on Microsoft Fabric — [\[Repo\]](#)

- Architected an end-to-end Microsoft Fabric pipeline to transform multi-source operational claims data into a governed, warehouse-ready format utilizing Data Vault 2.0 methodology.
- Orchestrated a five-stage T-SQL notebook pipeline encompassing Bronze ingestion, Silver standardization, Raw Vault loading, Gold mart construction, and final Warehouse publishing.
- Designed a robust Raw Vault architecture comprising Hubs, Links, and Satellites to map complex entity relationships across claims, policies, customers, vehicles, and payments.
- Delivered and validated three business-facing SQL warehouse tables (`wh_claim_360`, `wh_claim_history_timeline`, and `wh_fraud_support_case`) to enable optimized downstream analytics.

Smart Home Energy Management System — [\[Backend Repo\]](#) — [\[Demo\]](#)

- Architected a microservices-based smart home energy management platform to simulate and optimize real-time power interactions across solar, battery, grid, and home appliances.
- Engineered a multi-language backend featuring a Node.js/TypeScript simulation engine via Socket.IO, a Python Reinforcement Learning (PPO) optimization service, and a Go microservice (`gorilla/mux`) for RESTful device state management.
- Modeled complex energy consumption for 11 appliance types—including HVAC, water heaters, and EV charging—incorporating time-of-use pricing and thermal behavior to drive AI-based scheduling decisions.
- Built an interactive Vue.js dashboard with live charts to monitor the optimization loop, orchestrating the full end-to-end system deployment using Docker Compose.

Server Automation, Hardening & Observability — [\[Repo\]](#)

- Architected a fully automated infrastructure pipeline using Terraform and an 8-role Ansible project to provision and manage production Linux servers on Google Cloud Platform (GCP).
- Engineered zero-downtime blue-green deployments via Docker Compose, dynamically configuring Nginx reverse proxies and automating Let's Encrypt SSL/TLS renewals.
- Hardened system security by enforcing UFW default-deny policies, configuring fail2ban, establishing SSH key-only access, and provisioning a multi-peer WireGuard VPN.
- Integrated a comprehensive Prometheus and Grafana monitoring stack, ultimately reducing server setup time from hours of manual configuration to a repeatable 10-minute automated run.

AI Clinical Data Reconciliation Engine — [\[Repo\]](#) — [\[Live Demo\]](#)

- Architected a full-stack healthcare platform to reconcile conflicting medication records and validate patient EHR data quality across multiple dimensions.
- Engineered a Node.js/Express REST API integrated with the Google Gemini 2.5 Flash LLM to execute clinical reasoning, assess data plausibility, and generate weighted confidence scores.
- Hardened the live API for production by implementing rate limiting, API key authentication, health monitoring, and an SQLite caching layer that successfully reduced LLM costs by 60%.
- Containerized the React/Node system via Docker and deployed to Render, ensuring reliability through an automated Jest/Supertest suite covering auth, cache behavior, and rate-limit edge paths.